The University of Azad Jammu and Kashmir, Muzaffarabad



Digital Image Processing (DIP)
(Lab – 01)

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Roll no	**2024-SE-17**
Semester	**5th**
Batch	**2024-2028**
Lab	**01**
Submitted to	**Engr. Fahad Nisar**
Submission date	**Oct 23, 2026**

Department of Software Engineering

Step # 01: Python version Installation

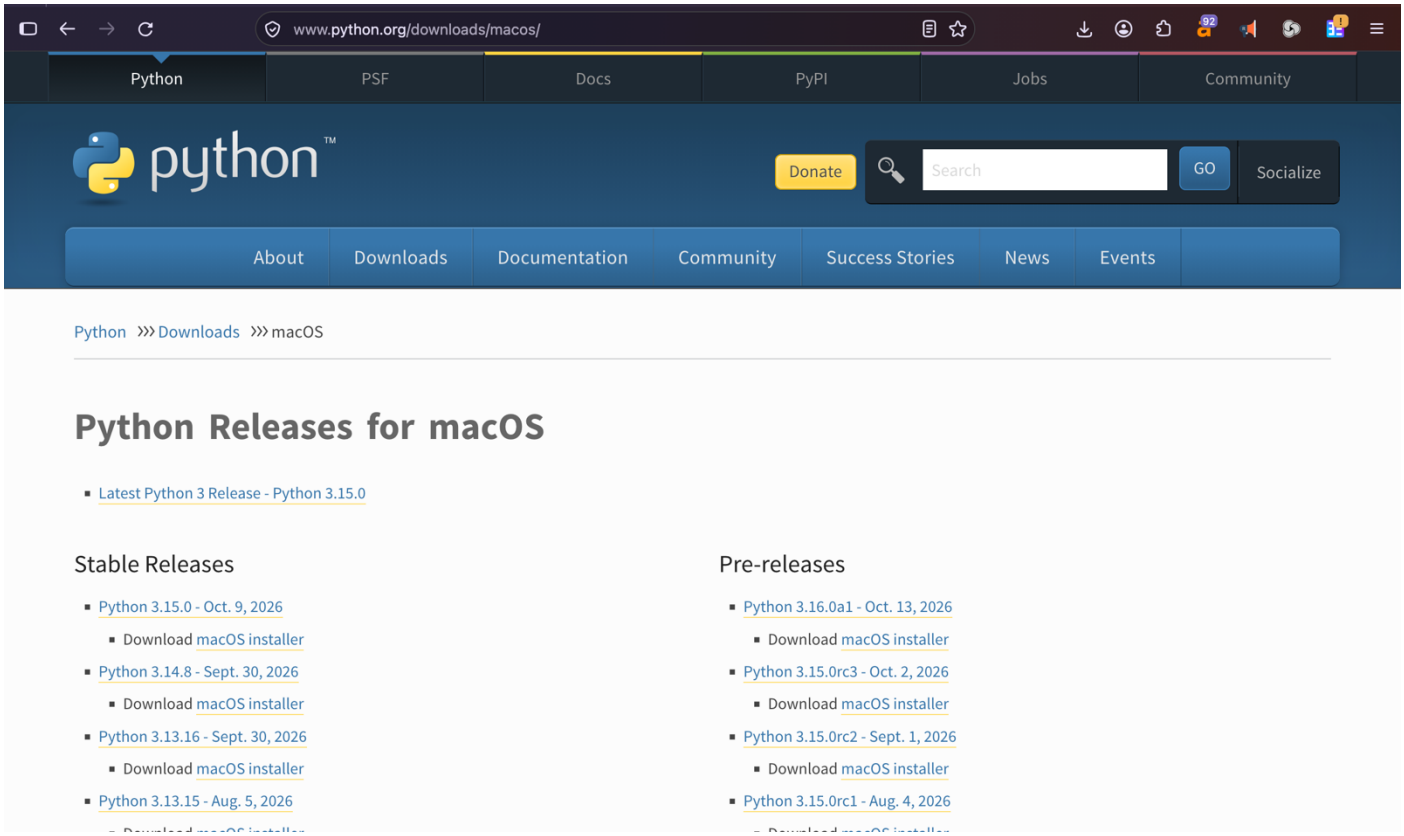
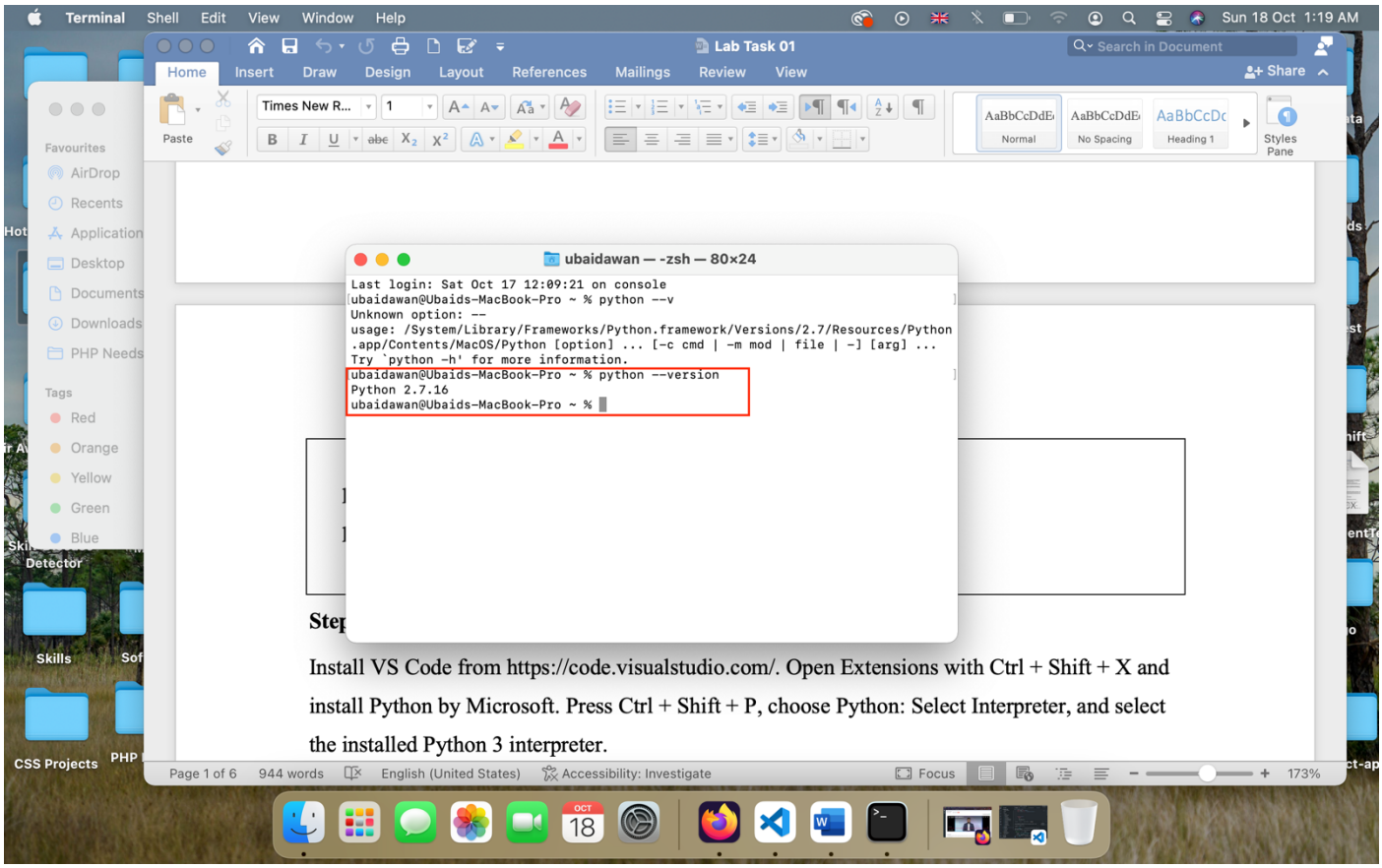
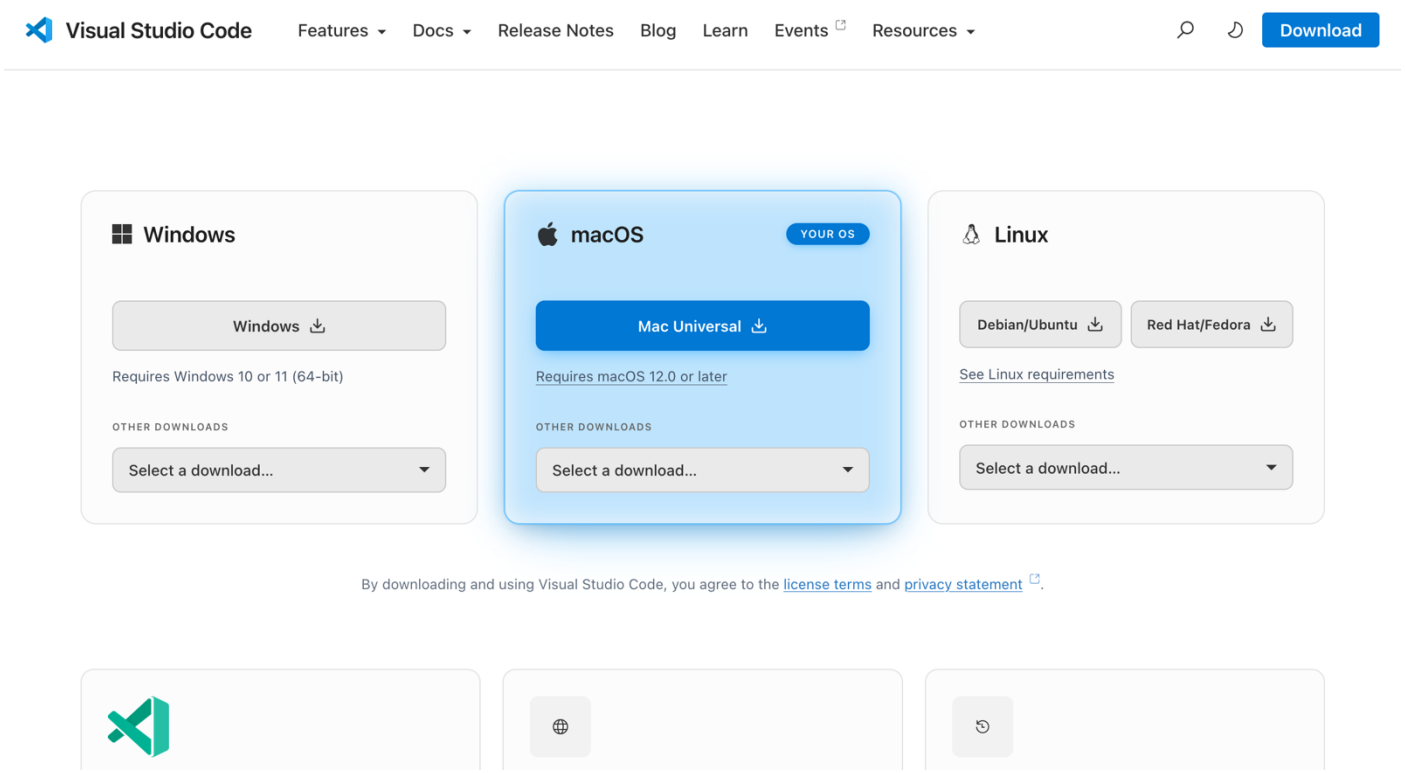


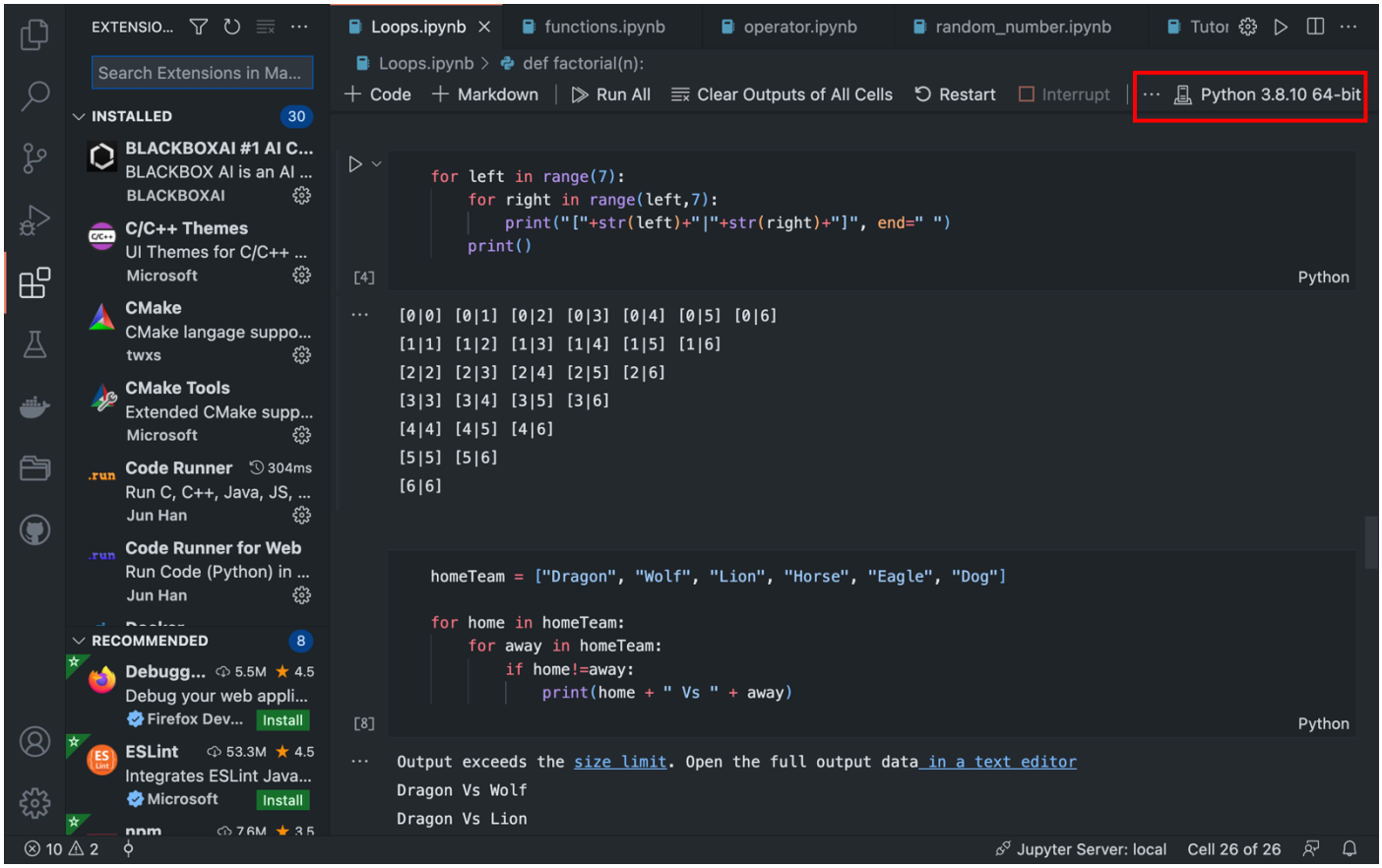
Figure 2Python setup download

Step #02: Python version in Cmd



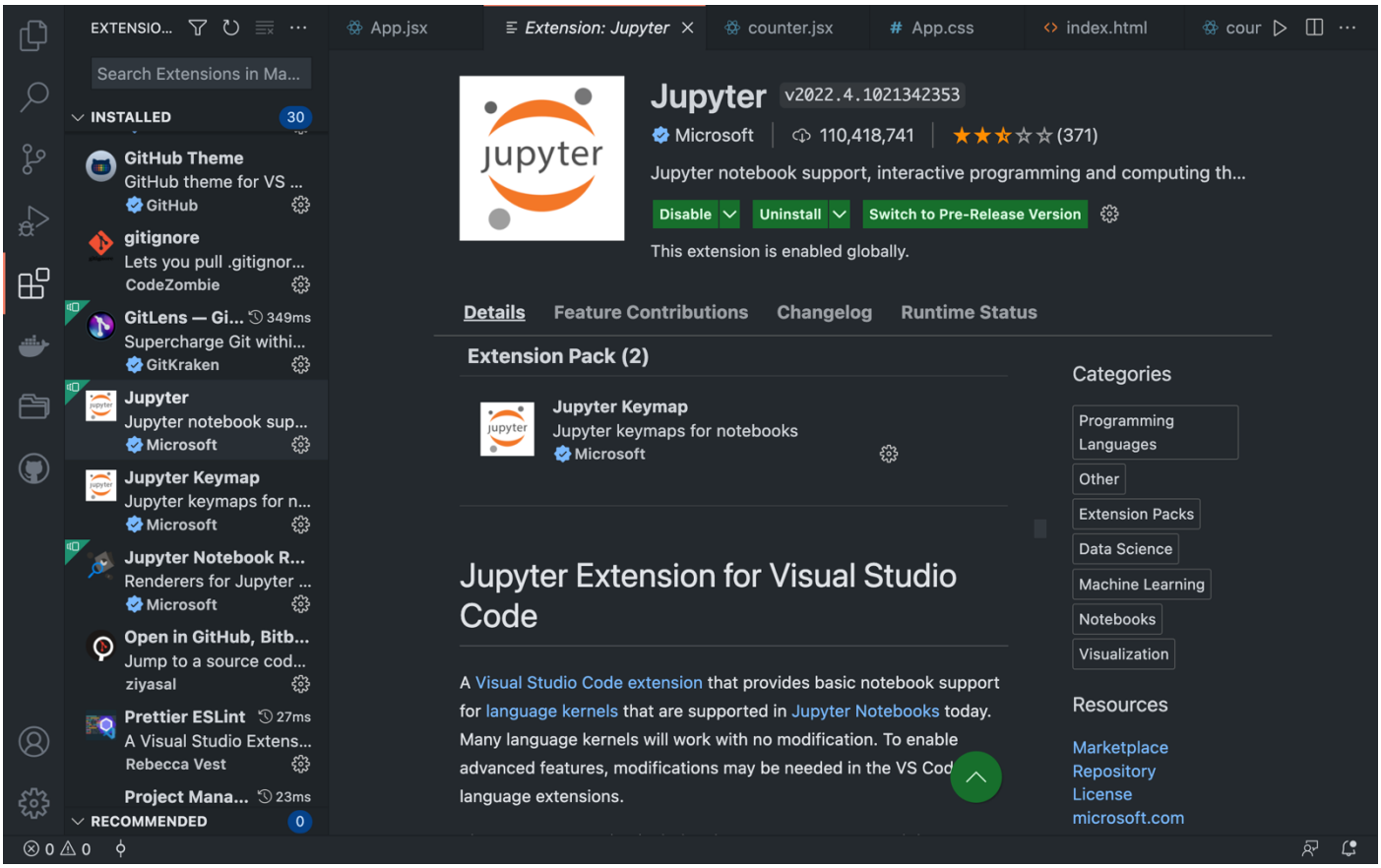
Step #03: Vs Code Installation



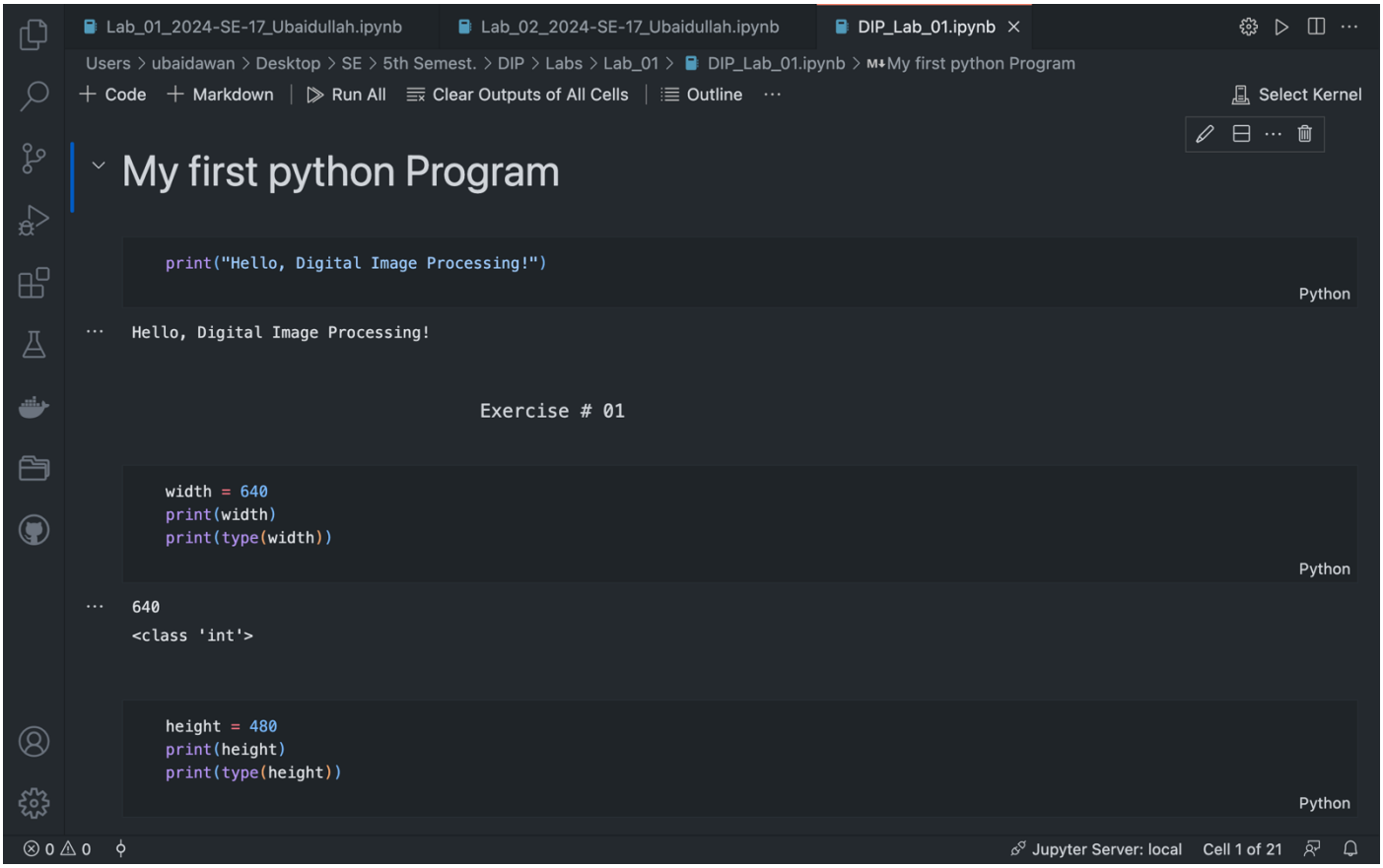


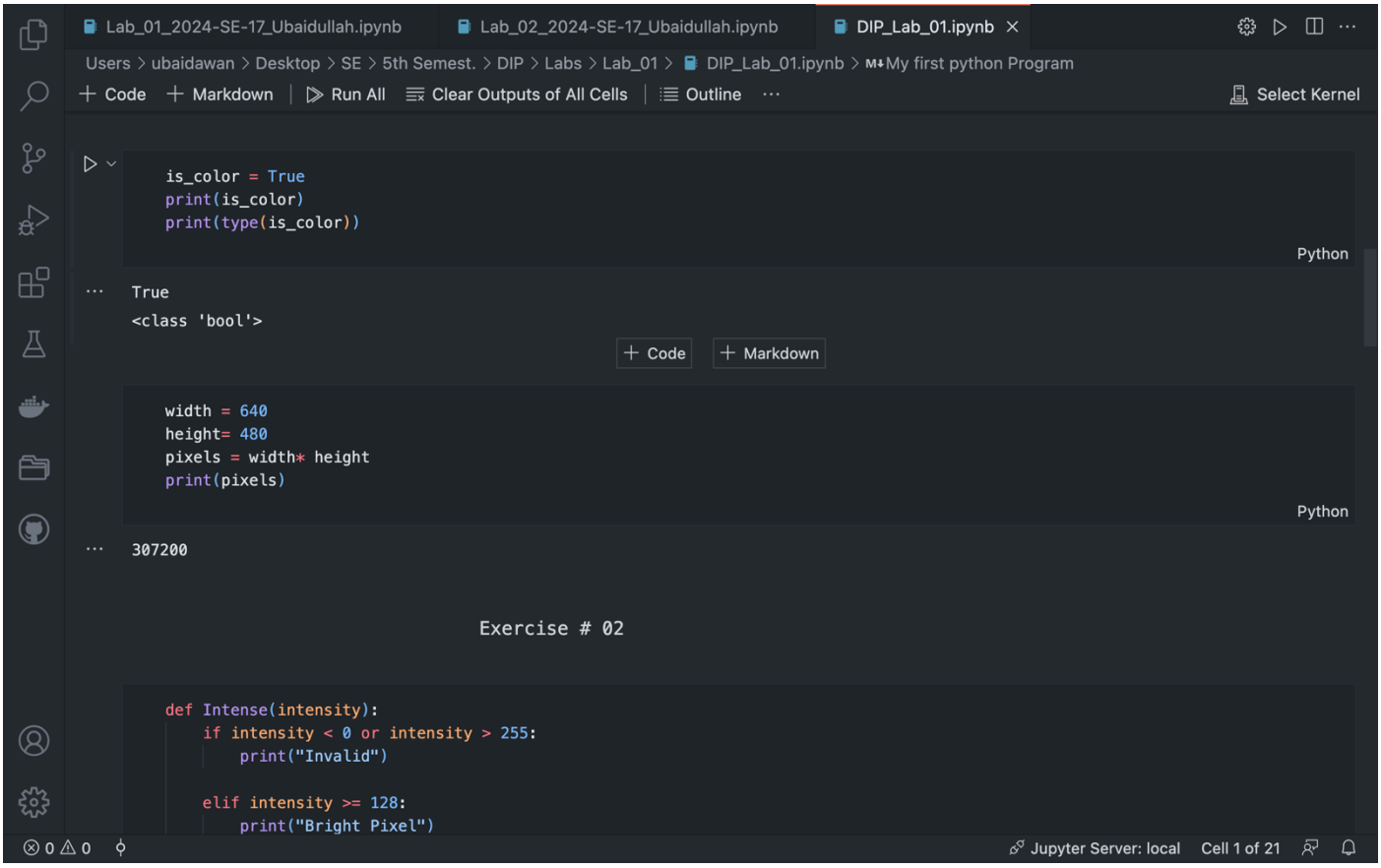
Step #04: Installing Extensions in Vscode

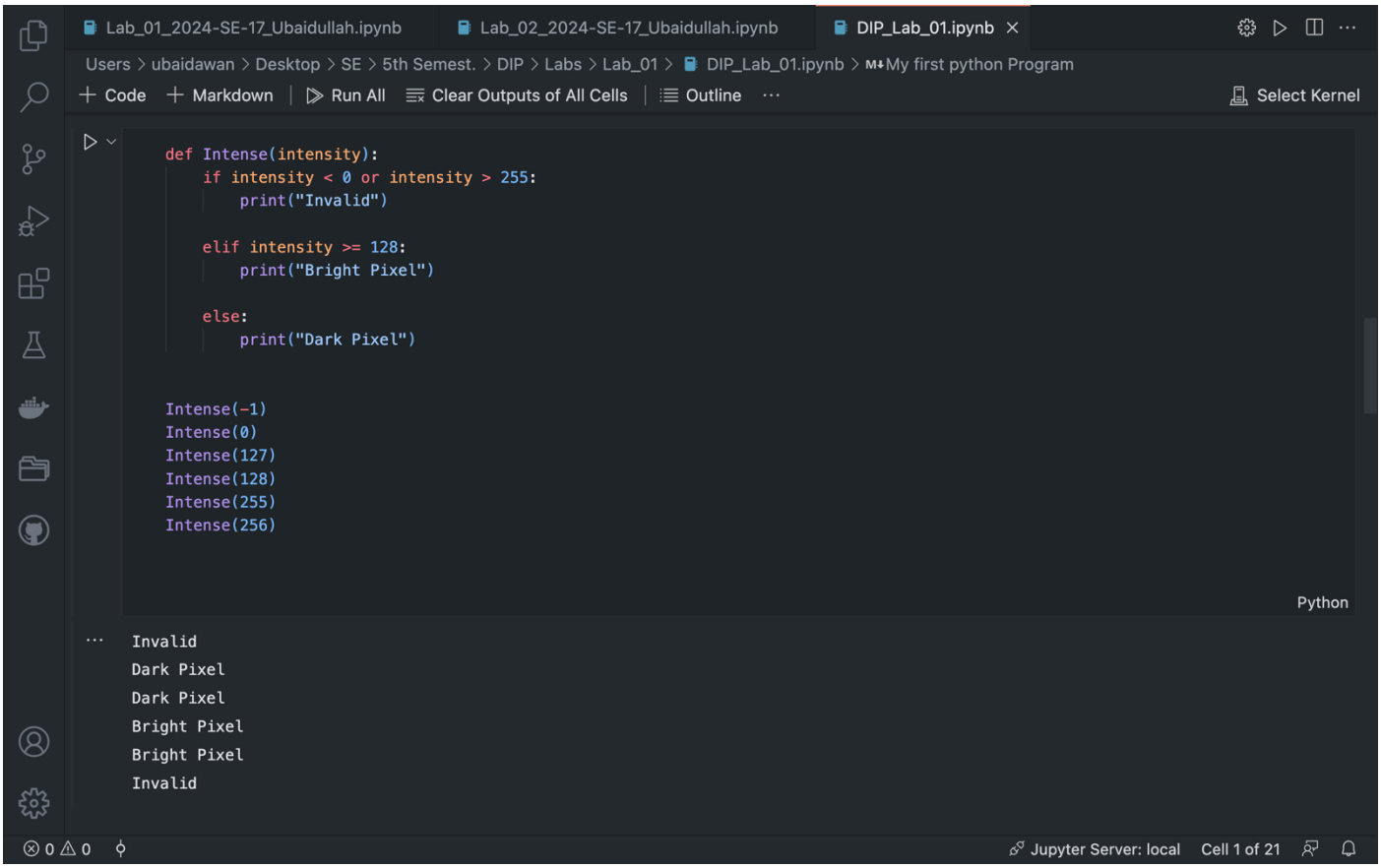


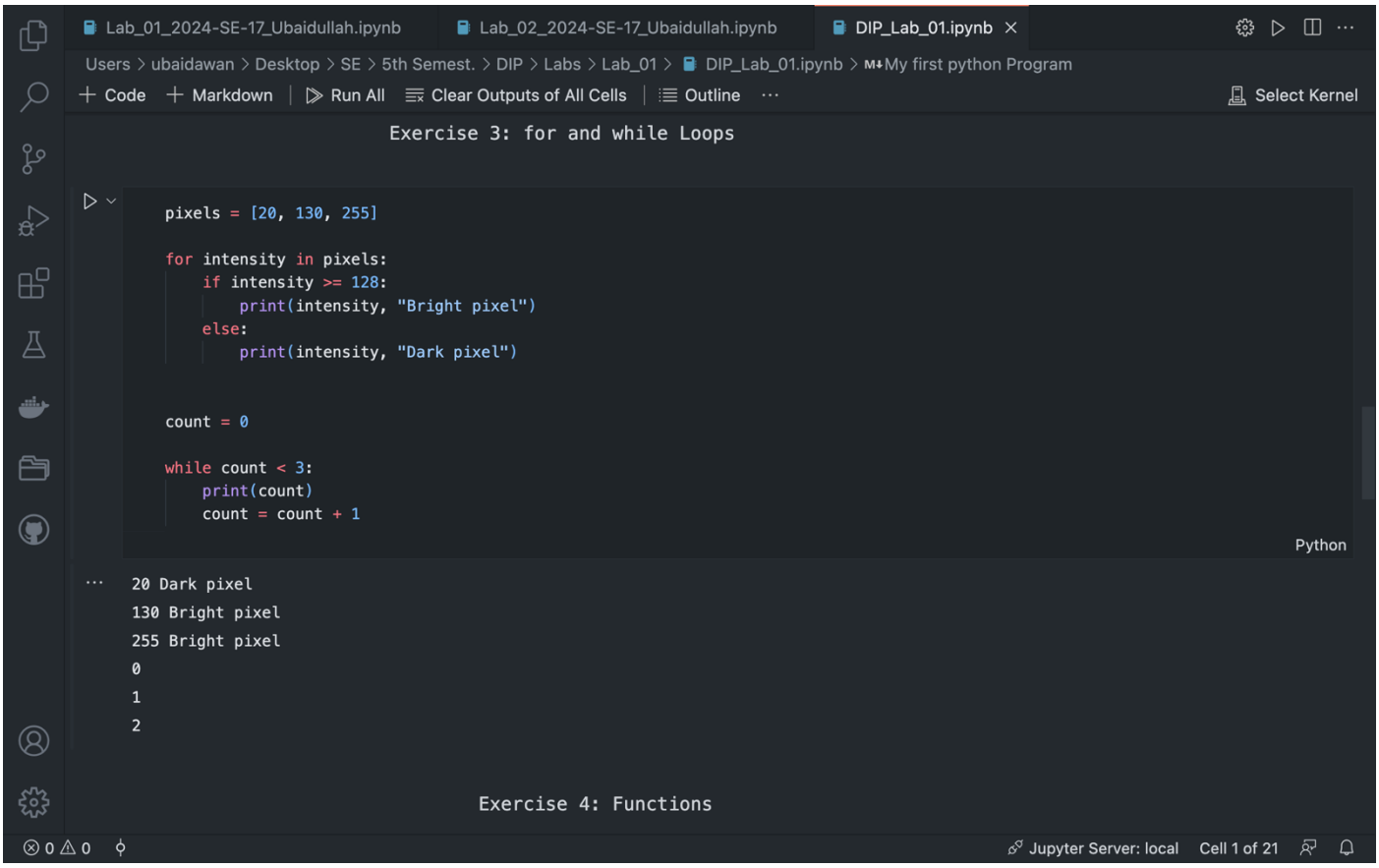

EXTENSIO...
Search Extensions in Ma...
INSTALLED 30
GitHub Theme
GitHub theme for VS ...
GitHub
gitignore
Lets you pull .gitignor...
CodeZombie
GitLens — Gi... 349ms
Supercharge Git withi...
GitKraken
Jupyter
Jupyter notebook sup...
Microsoft
Jupyter Keymap
Jupyter keymaps for n...
Microsoft
Jupyter Notebook R...
Renderers for Jupyter ...
Microsoft
Open in GitHub, Bitb...
Jump to a source cod...
ziyasal
Prettier ESLint 27ms
A Visual Studio Extens...
Rebecca Vest
Project Mana... 23ms
RECOMMENDED 0
App.jsx
Extension: Jupyter
counter.jsx
App.css
index.html
Jupyter v2022.4.1021342353
Microsoft 110,418,741 (371)
Jupyter notebook support, interactive programming and computing th...
Disable
Uninstall
Switch to Pre-Release Version
This extension is enabled globally.
Details
Feature Contributions
Changelog
Runtime Status
Extension Pack (2)
Jupyter Keymap
Jupyter keymaps for notebooks
Microsoft
Jupyter Extension for Visual Studio Code
A Visual Studio Code extension that provides basic notebook support for language kernels that are supported in Jupyter Notebooks today. Many language kernels will work with no modification. To enable advanced features, modifications may be needed in the VS Cod
language extensions.
Categories
Programming Languages
Other
Extension Packs
Data Science
Machine Learning
Notebooks
Visualization
Resources
Marketplace
Repository
License
microsoft.com

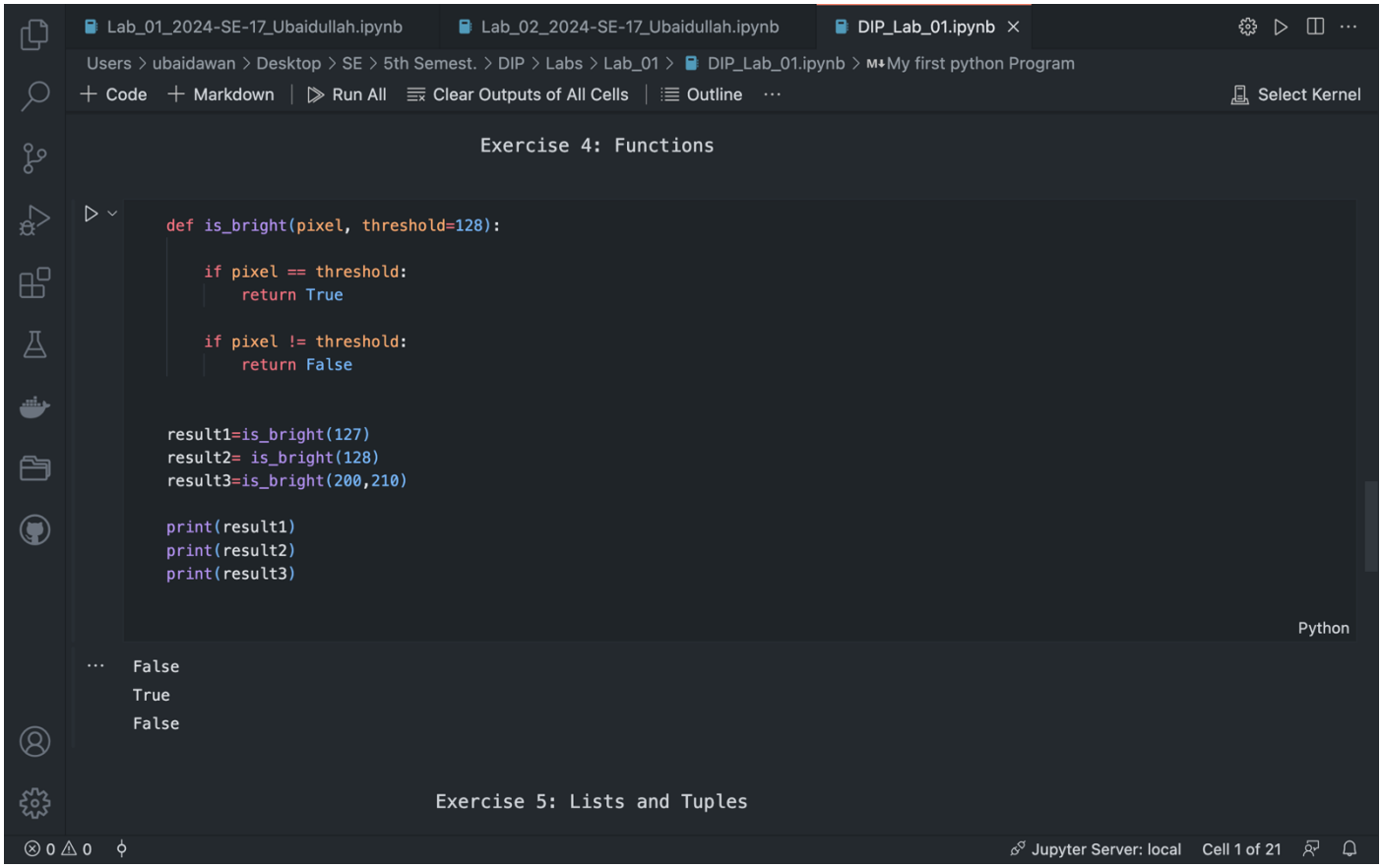
Step #05: Basic Exercises

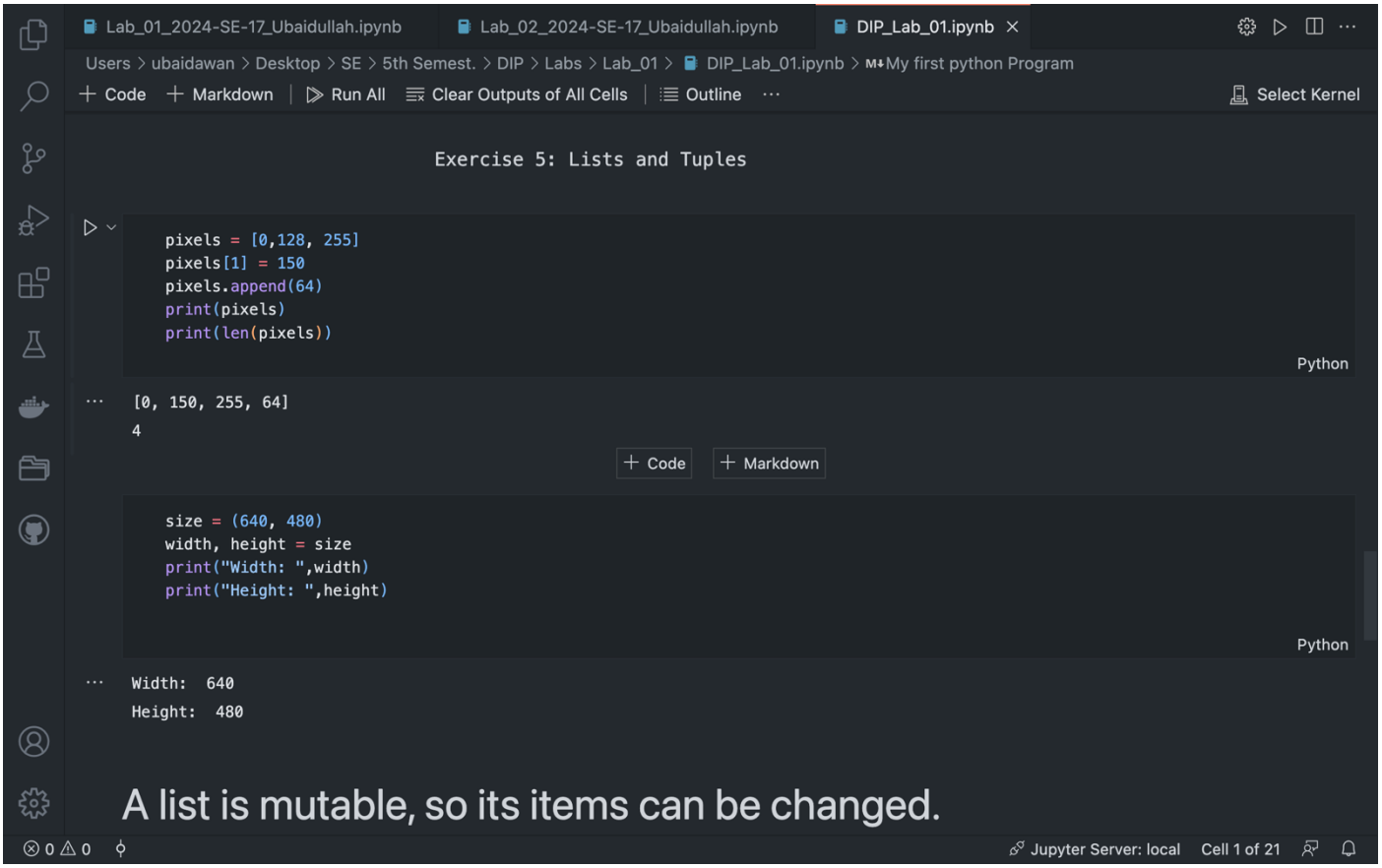


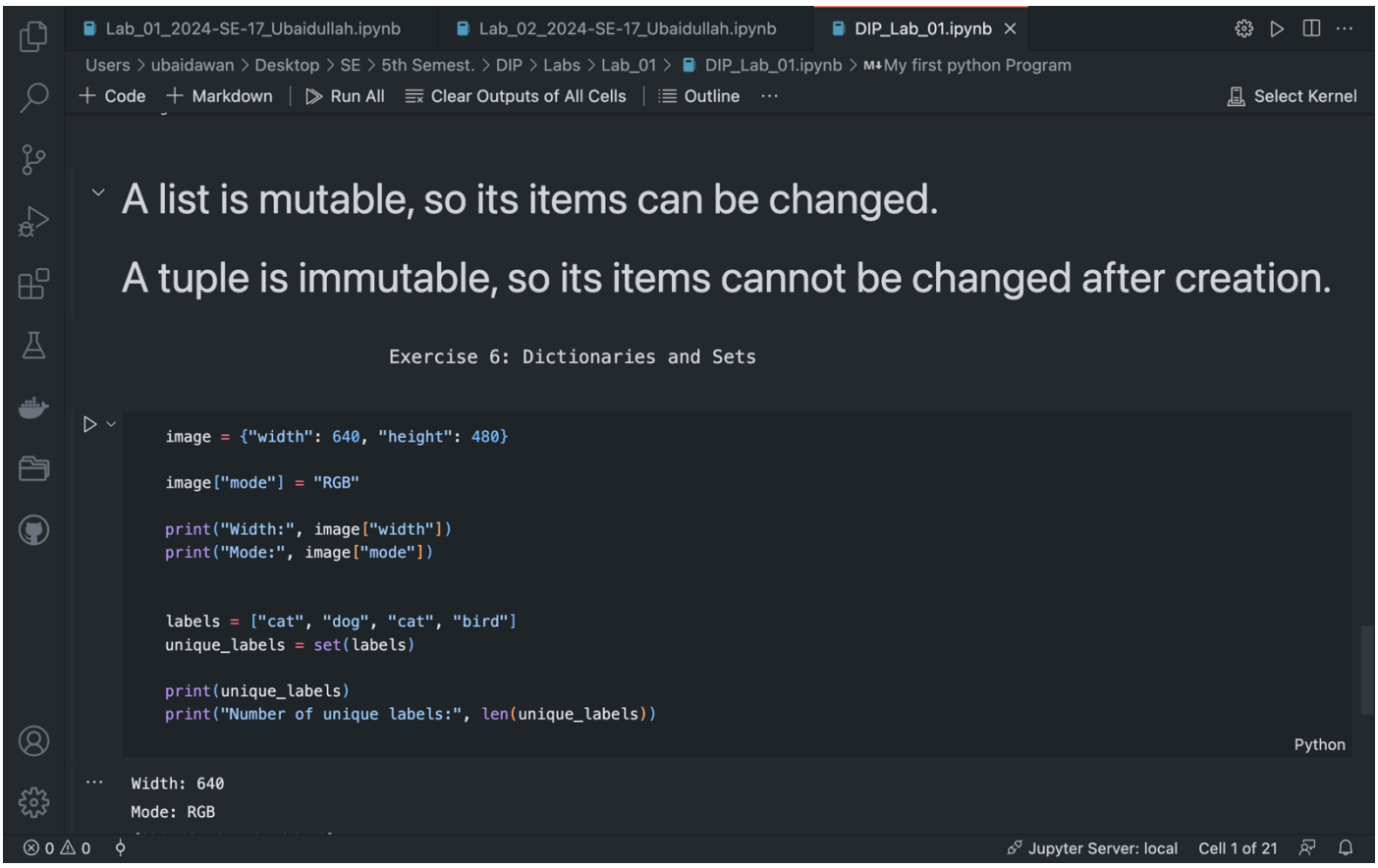


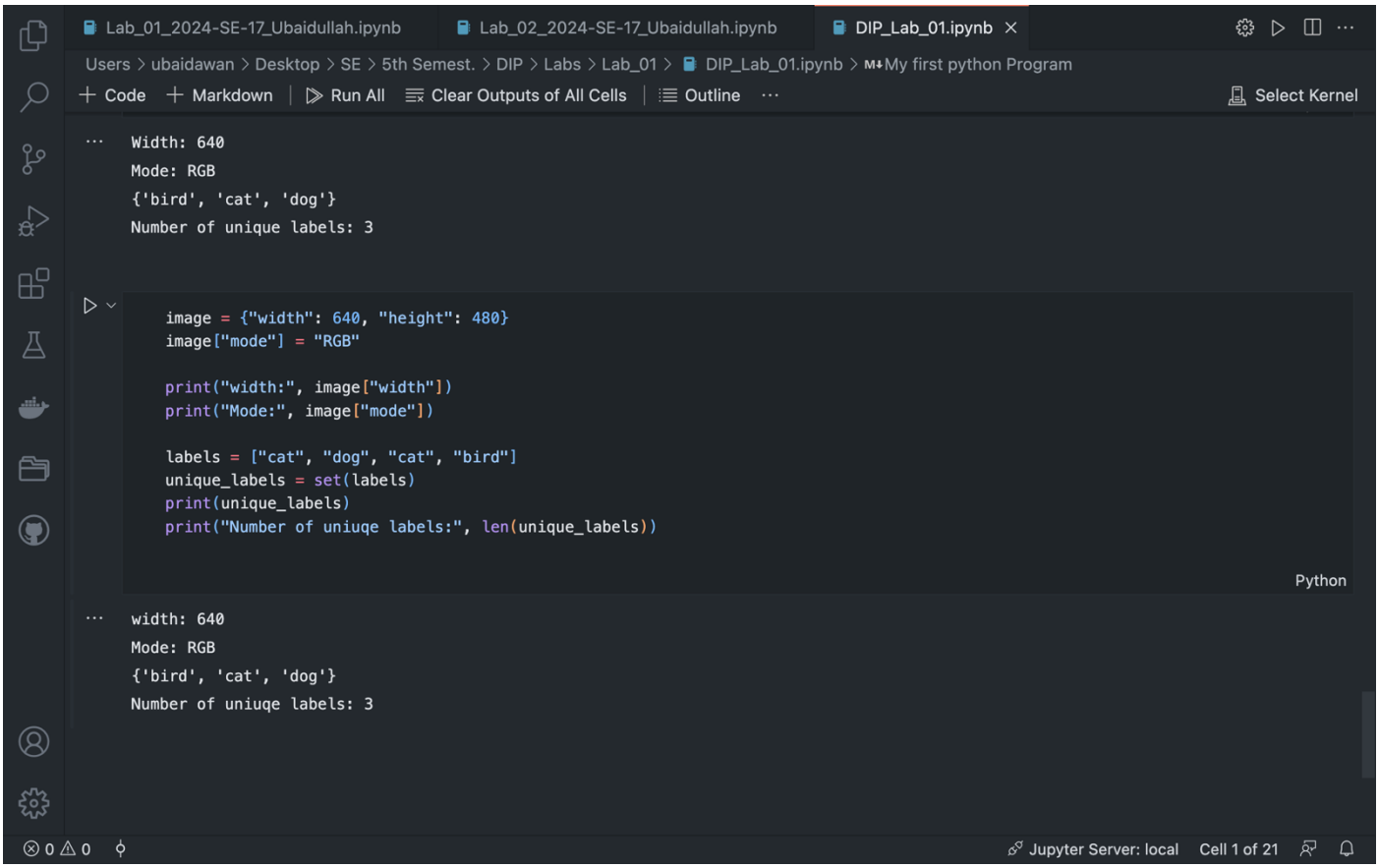

Lab_01_2024-SE-17_Ubaidullah.ipynb
Lab_02_2024-SE-17_Ubaidullah.ipynb
DIP_Lab_01.ipynb
Users > ubaidawan > Desktop > SE > 5th Semest. > DIP > Labs > Lab_01 > DIP_Lab_01.ipynb > My first python Program
Code
Markdown
Run All
Clear Outputs of All Cells
Outline
Select Kernel
def Intense(intensity):
if intensity < 0 or intensity > 255:
print("Invalid")
elif intensity >= 128:
print("Bright Pixel")
else:
print("Dark Pixel")
Intense(-1)
Intense(0)
Intense(127)
Intense(128)
Intense(255)
Intense(256)
Python
Invalid
Dark Pixel
Dark Pixel
Bright Pixel
Bright Pixel
Invalid
Jupyter Server: local
Cell 1 of 21


Lab_01_2024-SE-17_Ubaidullah.ipynb
Lab_02_2024-SE-17_Ubaidullah.ipynb
DIP_Lab_01.ipynb
Users > ubaidawan > Desktop > SE > 5th Semest. > DIP > Labs > Lab_01 > DIP_Lab_01.ipynb > My first python Program
Code
Markdown
Run All
Clear Outputs of All Cells
Outline
Select Kernel
Exercise 3: for and while Loops
pixels = [20, 130, 255]
for intensity in pixels:
if intensity >= 128:
print(intensity, "Bright pixel")
else:
print(intensity, "Dark pixel")
count = 0
while count < 3:
print(count)
count = count + 1
Python
20 Dark pixel
130 Bright pixel
255 Bright pixel
0
1
2
Exercise 4: Functions
Jupyter Server: local
Cell 1 of 21


Lab_01_2024-SE-17_Ubaidullah.ipynb
Lab_02_2024-SE-17_Ubaidullah.ipynb
DIP_Lab_01.ipynb
Users > ubaidawan > Desktop > SE > 5th Semest. > DIP > Labs > Lab_01 > DIP_Lab_01.ipynb > My first python Program
Code
Markdown
Run All
Clear Outputs of All Cells
Outline
Select Kernel
Exercise 4: Functions
def is_bright(pixel, threshold=128):
if pixel == threshold:
return True
if pixel != threshold:
return False
result1=is_bright(127)
result2= is_bright(128)
result3=is_bright(200,210)
print(result1)
print(result2)
print(result3)
Python
False
True
False
Exercise 5: Lists and Tuples
Jupyter Server: local
Cell 1 of 21

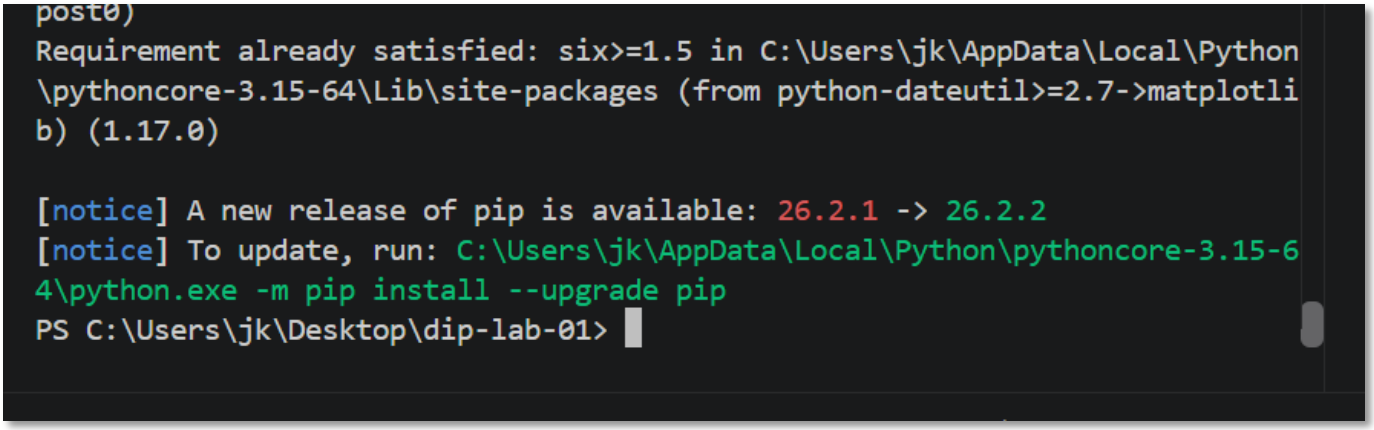

Lab_01_2024-SE-17_Ubaidullah.ipynb
Lab_02_2024-SE-17_Ubaidullah.ipynb
DIP_Lab_01.ipynb
Users > ubaidawan > Desktop > SE > 5th Semest. > DIP > Labs > Lab_01 > DIP_Lab_01.ipynb > My first python Program
Code
Markdown
Run All
Clear Outputs of All Cells
Outline
Select Kernel
Exercise 5: Lists and Tuples
pixels = [0,128, 255]
pixels[1] = 150
pixels.append(64)
print(pixels)
print(len(pixels))
Python
[0, 150, 255, 64]
4
Code
Markdown
size = (640, 480)
width, height = size
print("Width: ",width)
print("Height: ",height)
Python
Width: 640
Height: 480
A list is mutable, so its items can be changed.
Jupyter Server: local
Cell 1 of 21

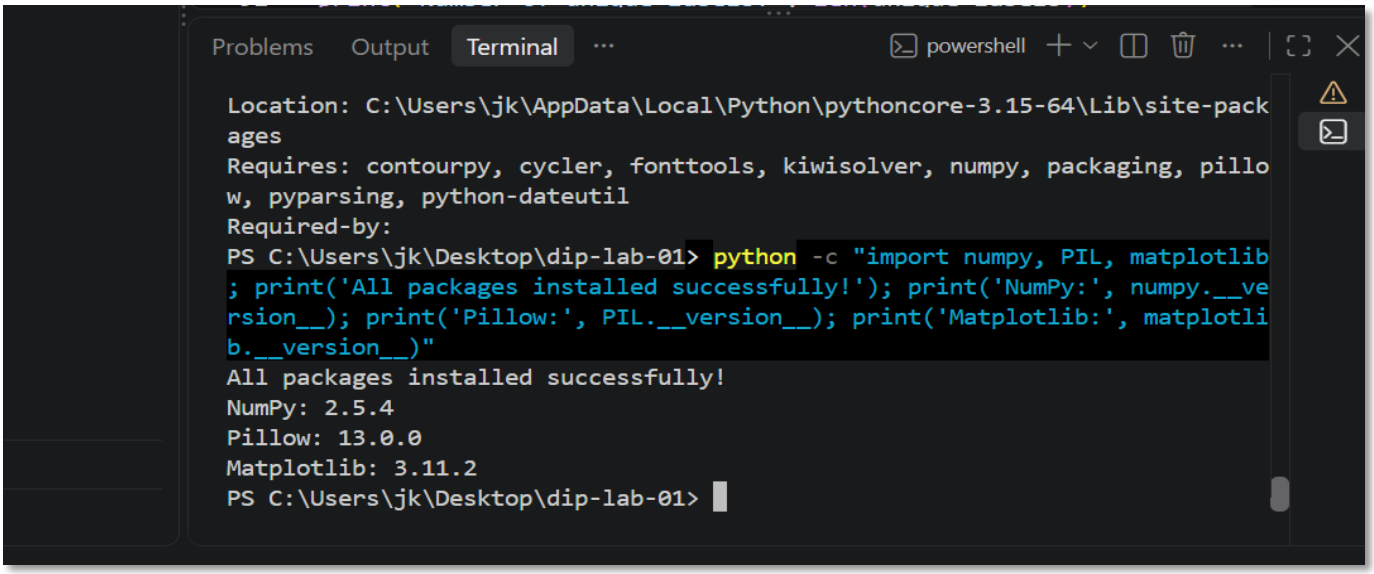

Lab_01_2024-SE-17_Ubaidullah.ipynb
Lab_02_2024-SE-17_Ubaidullah.ipynb
DIP_Lab_01.ipynb
Users > ubaidawan > Desktop > SE > 5th Semest. > DIP > Labs > Lab_01 > DIP_Lab_01.ipynb > My first python Program
Code
Markdown
Run All
Clear Outputs of All Cells
Outline
Select Kernel
A list is mutable, so its items can be changed.
A tuple is immutable, so its items cannot be changed after creation.
Exercise 6: Dictionaries and Sets
image = {"width": 640, "height": 480}
image["mode"] = "RGB"
print("Width:", image["width"])
print("Mode:", image["mode"])
labels = ["cat", "dog", "cat", "bird"]
unique_labels = set(labels)
print(unique_labels)
print("Number of unique labels:", len(unique_labels))
Python
Width: 640
Mode: RGB
Jupyter Server: local
Cell 1 of 21


Lab_01_2024-SE-17_Ubaidullah.ipynb
Lab_02_2024-SE-17_Ubaidullah.ipynb
DIP_Lab_01.ipynb
Users > ubaidawan > Desktop > SE > 5th Semest. > DIP > Labs > Lab_01 > DIP_Lab_01.ipynb > My first python Program
Code
Markdown
Run All
Clear Outputs of All Cells
Outline
Select Kernel
Width: 640
Mode: RGB
{'bird', 'cat', 'dog'}
Number of unique labels: 3
image = {"width": 640, "height": 480}
image["mode"] = "RGB"
print("width:", image["width"])
print("Mode:", image["mode"])
labels = ["cat", "dog", "cat", "bird"]
unique_labels = set(labels)
print(unique_labels)
print("Number of uniuqe labels:", len(unique_labels))
Python
width: 640
Mode: RGB
{'bird', 'cat', 'dog'}
Number of uniuqe labels: 3
Jupyter Server: local
Cell 1 of 21

Step #06: Installing the required packages

Command: `python -m pip install numpy pillow matplotlib`





Command: `python -m pip show numpy pillow matplotlib`

This displays the name, version, and installation location of all three packages.

```
Name: pillow
Version: 13.0.0
Summary: Python Imaging Library (fork)
Home-page: https://python-pillow.org
PS C:\Users\HP\OneDrive\Desktop\dip-lab-01> python -m pip show numpy pillow matplotlib | more
Name: numpy
Version: 2.5.4
Summary: Fundamental package for array computing in Python
Home-page: https://numpy.org
Author: Travis E. Oliphant et al.
Author-email:
License-Expression: BSD-3-Clause AND 0BSD AND MIT AND Zlib AND CC0-1.0
Location: C:\Users\HP\AppData\Roaming\Python\Python314\site-packages
Requires:
Required-by: contourpy, matplotlib, pandas
Name: matplotlib
Version: 3.11.2
Summary: Python plotting package
Home-page: https://matplotlib.org
Author: John D. Hunter, Michael Droettboom
Author-email: Unknown <matplotlib-users@python.org>
License: License agreement for matplotlib versions 1.3.0 and later
```

Step #08: Writing and verifying the import statements

lab01_imports.py

```
import numpy as np
from PIL import Image
import matplotlib.pyplot as plt
```

lab01_imports.py

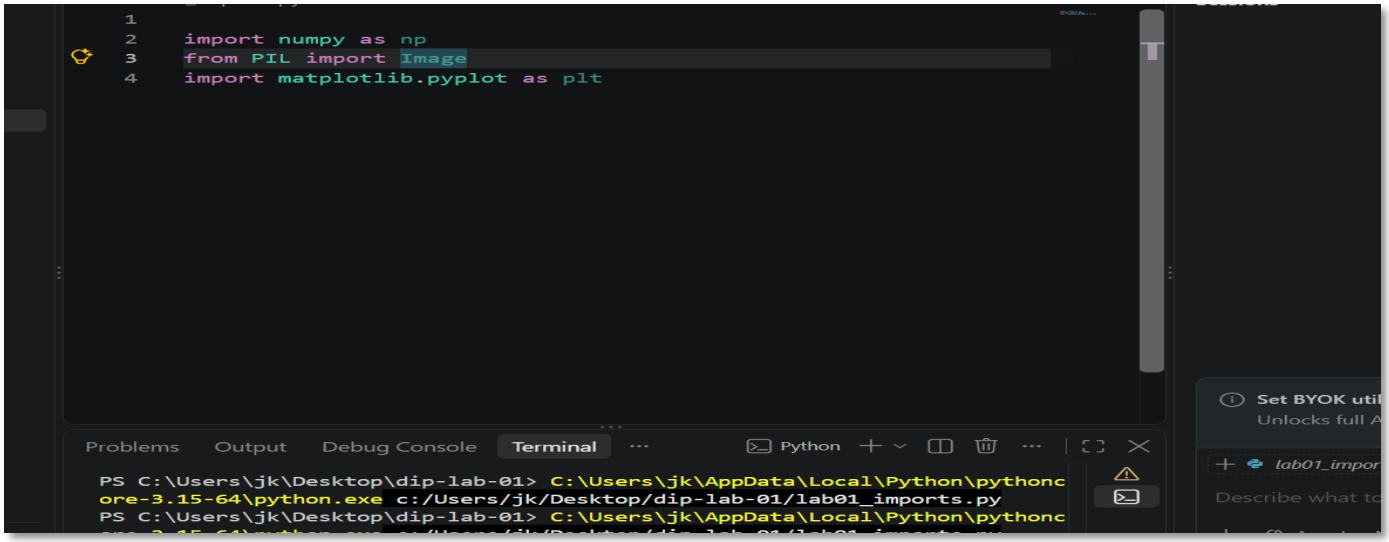
```
"""
================================================================================
File:           lab01_imports.py
Description:    Lab Task 03 - Package Import Verification
Course:         Digital Image Processing Lab
Author:         Sheikh Haseeb-ur-Rehman
Roll Number:    2024-SE-31
Instructor:     Engr. Fahad Nisar
================================================================================
"""

import numpy as np
from PIL import Image
import matplotlib.pyplot as plt
```

Problems Debug Console Terminal Ports Output

```
PS C:\Users\HP\OneDrive\Desktop\dip-lab-01> python lab01_imports.py
PS C:\Users\HP\OneDrive\Desktop\dip-lab-01>
```

Command: `python lab01_imports.py`



The program finished without errors and returned to the command prompt. No output is printed because the file contains only import statements.

Python versus pip, and the purpose of the imports

Python is the programming language and interpreter that runs our .py files, while pip is the package manager that comes with Python and is used to download and install extra libraries such as NumPy, Pillow and Matplotlib. Running pip as "python -m pip" makes sure the packages are installed for the same Python interpreter that will run the code.

The import statements load those installed libraries into a program so their features can be used. NumPy (imported as np) provides fast numerical arrays, which is how images are stored as pixel values; Pillow (from PIL import Image) opens, saves and edits image files; and Matplotlib (imported as plt) displays images and draws plots. Running lab01_imports.py without errors confirms that all three packages are correctly installed.